# Activity 1 - before.pdf

First attempt (weak prompt). NotebookLM response with sources P1-P5 selected.

Prompt used: *Summarize these five papers and tell me their methodologies and results.*

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Here is a structured overview of the five papers, covering the core summary, methodology, and experimental results for each.

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## 1. "Attention Is All You Need" (Vaswani et al., 2017)

*Source:* `1706.03762v7.pdf`

- **Overview:** This foundational paper introduces the **Transformer**, a sequence transduction architecture that entirely dispenses with recurrence (RNNs) and convolutions. Instead, it relies solely on attention mechanisms to model global dependencies between input and output representations.
- **Methodology:**
  - **Encoder-Decoder Architecture:** Both encoder and decoder stacks consist of N = 6 identical layers.
  - **Sub-layers:** Each encoder layer contains two sub-layers: Multi-Head Self-Attention and a position-wise fully connected feed-forward network, wrapped in residual connections and layer normalization (LayerNorm(x + Sublayer(x))) with model dimension d_model = 512. The decoder contains a third sub-layer performing multi-head attention over the encoder output, alongside masked self-attention to prevent positions from attending to future tokens.
  - **Attention Formulation:** Employs **Scaled Dot-Product Attention**, defined as softmax((QK^T)/(sqrt(d_k)))V, and **Multi-Head Attention** using h = 8 parallel heads with dimensions d_k = d_v = 64.
  - **Training Setup:** Uses sinusoidal positional encodings, byte-pair/word-piece tokenization, the Adam optimizer with warmup and decay schedules, dropout (P_drop = 0.1), and label smoothing (eps_ls = 0.1).
- **Results:**
  - **Machine Translation:** On the WMT 2014 English-to-German task, the Transformer (big) achieved a state-of-the-art score of **28.4 BLEU**, surpassing previous models and ensembles by more than 2.0 BLEU. On WMT 2014 English-to-French, it established a single-model state-of-the-art of **41.8 BLEU** (and 41.0 on newstest2014) after training for only 3.5 days on 8 GPUs.
  - **Efficiency:** Training was completed in a fraction of the time and computational cost of previous recurrent and convolutional architectures.
  - **Generalization:** Applied successfully to English constituency parsing on WSJ Section 23, achieving **91.3 F1** discriminatively (WSJ only) and **92.7 F1** semi-supervised, outperforming the Berkeley Parser.

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## 2. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" (Lewis et al., 2020)

*Source:* `2005.11401v4.pdf`

- **Overview:** This paper presents **Retrieval-Augmented Generation (RAG)**, a hybrid framework combining pre-trained parametric memory (a sequence-to-sequence neural network) with an explicit non-parametric memory (a dense vector index of Wikipedia) accessed via a neural retriever.
- **Methodology:**

    - **Architecture Components:** Combines Dense Passage Retriever (DPR)—a BERT-base bi-encoder (d(z) and q(x)) over 21 million 100-word Wikipedia passages indexed via FAISS Maximum Inner Product Search (MIPS)—with a BART-large generator (400M parameters).
    - **Formulations:** Introduces two marginalization strategies over latent retrieved documents:
        - **RAG-Sequence:** Uses the same retrieved passage across all tokens of the generated sequence, marginalizing over documents at the sequence level.
        - **RAG-Token:** Draws and marginalizes different documents token-by-token, enabling information synthesis across distinct sources.
    - **Training & Optimization:** Jointly trains the query encoder and BART generator end-to-end to minimize negative marginal log-likelihood without direct supervision on which document to retrieve, keeping the document index fixed.
    - **Knowledge Hot-Swapping:** Demonstrates that world knowledge can be updated at test time simply by swapping the non-parametric document index without retraining model weights.
- **Results:**
    - **Open-Domain QA:** Set state-of-the-art Exact Match (EM) scores across multiple benchmarks: **44.5 EM** on Natural Questions (RAG-Seq), **45.5 EM** on WebQuestions (RAG-Token), **52.2 EM** on CuratedTrec (RAG-Seq), and **68.0 EM** on the TriviaQA Wiki split. RAG outperformed closed-book models like T5-11B and extractive models like DPR and REALM. It even generated correct answers in 11.8% of Natural Questions cases where the answer was absent from the retrieved passages.
    - **Text Generation & Fact Checking:** Outperformed BART on MS-MARCO abstractive QA by 2.6 BLEU and 2.6 ROUGE-L. On Jeopardy question generation, human evaluators found RAG factually accurate in 42.7% of cases versus 7.1% for BART. On FEVER fact verification, it reached **74.5% label accuracy** (3-way) without retrieval supervision, within 4.3% of specialized pipeline models.

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## 3. "Language Models are Few-Shot Learners" (Brown et al., 2020)

*Source:* `2005.14165v4.pdf`

- **Overview:** Demonstrates that scaling autoregressive language models dramatically enhances **in-context learning**, allowing models to solve downstream language tasks without any fine-tuning or gradient updates. This work introduces **GPT-3**.
- **Methodology:**
    - **Model Architecture & Scale:** Autoregressive Transformer decoders with alternating dense and locally banded sparse attention patterns. Evaluates 8 model sizes spanning 125M to 175B parameters. GPT-3 175B features 96 layers, 96 attention heads, and a hidden dimension of 12,288, trained on 300 billion tokens.
    - **Pre-training Data:** Filtered Common Crawl, WebText2, Books1, Books2, and Wikipedia.
    - **Evaluation Conditions:** Evaluated strictly via text prompts across three settings:
        - **Zero-shot:** Task description only.
        - **One-shot:** Task description with exactly one demonstration.
        - **Few-shot:** Task description with 10 to 100 demonstrations fitting the 2,048-token context window.
- **Results:**
    - **Scaling Trends:** Performance scales predictably as a power-law function of compute and parameters across 42 accuracy benchmarks, with few-shot performance gaining the most from increased scale.
    - **Language Modeling & QA:** Set a zero-shot perplexity record on Penn Treebank (**20.50**). On LAMBADA, few-shot accuracy reached **86.4%**, an 18% improvement over the prior state of the art.

In closed-book QA, GPT-3 reached **71.2%** on TriviaQA few-shot (matching or exceeding fine-tuned open-domain models), **41.5%** on WebQuestions, and **29.9%** on Natural Questions.

- **Translation & Reasoning:** Outperformed prior unsupervised neural machine translation by up to 5 BLEU when translating into English (e.g., 40.6 BLEU on De->En and 39.2 BLEU on Fr->En). On PIQA commonsense reasoning, it achieved state-of-the-art accuracy of **82.8%**. On SuperGLUE, few-shot GPT-3 achieved an aggregate score of **71.8**, matching fine-tuned BERT-Large.
- **Novel & Synthetic Tasks:** Showed emergent abilities on word unscrambling (up to 67.2% on random insertion), 3-digit arithmetic, and synthetic news generation where humans distinguished GPT-3 articles from real ones with only ~52% accuracy.
- **Weaknesses:** Lagged behind fine-tuned systems on Natural Language Inference (ANLI, RTE) and certain bidirectional reading tasks (WiC, DROP, QuAC).

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## 4. "ReAct: Synergizing Reasoning and Acting in Language Models" (Yao et al., 2022)

*Source:* `2210.03629v3.pdf`

- **Overview:** Proposes **ReAct**, a prompting paradigm that prompts LLMs to generate **verbal reasoning traces and domain-specific actions in an interleaved fashion**. Reasoning traces help the model maintain and adjust plans, while actions allow it to interact with external environments to gather grounded facts.
- **Methodology:**
  - **Augmented Action Space:** Expands the action space to A' = A U L, where L represents natural language thoughts. Thoughts do not affect the external environment but update the internal context memory to guide subsequent actions and inferences.
  - **Trajectory Structure:** Prompts models with few-shot examples structured as repeating sequences of *Thought -> Action -> Observation*. Thoughts serve to decompose goals, track progress, inject commonsense, extract information from observations, and handle exceptions.
  - **Model Setups:** Tested with PaLM-540B and GPT-3 (`text-davinci-002`) using 1 to 6 exemplars, as well as fine-tuning PaLM-8B and 62B on 3,000 self-generated ReAct trajectories.
  - **Combination with Chain-of-Thought:** Integrates ReAct with Chain-of-Thought Self-Consistency (CoT-SC) by switching between external retrieval and internal reasoning when confidence or steps fail (e.g., ReAct -> CoT-SC or CoT-SC -> ReAct).
- **Results:**
  - **Knowledge Reasoning:** On HotpotQA multi-hop QA, ReAct reached **27.4 EM** (outperforming Act-only at 25.7) and reduced hallucinated reasoning traces to **0%** (compared to 56% in CoT). Combining ReAct -> CoT-SC reached **35.1 EM**. On FEVER fact verification, ReAct achieved **60.9% accuracy** (beating CoT at 56.3%), while CoT-SC -> ReAct reached **64.6%**.
  - **Decision Making Tasks:**
    - On **ALFWorld** (simulated household text game), few-shot ReAct achieved a **40.0% success rate** with PaLM-540B (and **78.4%** with GPT-3), dramatically outperforming imitation learning (29.1%) and reinforcement learning (28.7%), as well as Inner Monologue prompting (71% vs. 53%).
    - On **WebShop** (interactive shopping environment with 1.18M products), ReAct achieved an absolute **10% improvement** in success rate over prior imitation and reinforcement learning baselines (40.0% vs. 28.7–29.1%).
  - **Interpretability:** Provides human-interpretable trajectories that can be diagnosed and edited in real time to steer agent behavior.

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## 5. "SELF-REFINE: Iterative Refinement with Self-Feedback" (Madaan et al., 2023)

*Source:* `2303.17651v2.pdf`

- **Overview:** Introduces **SELF-REFINE**, a framework that enables a single pre-trained language model to iteratively improve its own outputs using self-generated natural language feedback and self-refinement, operating without supervised data, retraining, or reinforcement learning.
- **Methodology:**
  - **Iterative Feedback Loop:**
  1. *Initial Generation:* Base LLM M produces candidate output y_0 for input x.
  2. *Feedback:* The same model M evaluates y_t along multi-aspect, task-specific rubrics (e.g., efficiency, readability, clarity) and generates detailed natural language critique fb_t.
  3. *Refinement:* The same model M uses x, y_t, and fb_t to produce an updated output y_t+1.
     - Steps 2 and 3 repeat iteratively (up to a maximum of 4 iterations or until stopping criteria are met).
  - **Setup & Evaluated Models:** Tested across four frozen LLMs—GPT-3.5 (`text-davinci-003`), ChatGPT (`gpt-3.5-turbo`), GPT-4, and Codex (`code-davinci-002`)—prompted with few-shot examples.
  - **Evaluation Domains:** Evaluated on 7 generation tasks spanning text and code: Dialogue Response Generation, Code Optimization, Code Readability Improvement, Math Reasoning (GSM-8k), Sentiment Reversal, Acronym Generation, and Constrained Generation (CommonGen-Hard).
- **Results:**
  - **Overall Gains:** Across all evaluated tasks, SELF-REFINE improved performance over direct single-pass generation by **~20% absolute on average** across automatic metrics and human evaluations.
  - **Selected Task Results:**
    - *Dialogue Response Generation:* GPT-4 preference rate increased from 25.4% to **74.6%** (+49.2%).
    - *Sentiment Reversal:* GPT-4 preference increased from 3.8% to **36.2%** (+32.4%), and ChatGPT preference increased from 11.4% to **43.2%** (+31.8%).
    - *Code Optimization:* Programs successfully optimized on the PIE dataset rose from 27.3% to **36.0%** (+8.7%) for GPT-4, and improved up to 13% for Codex.
    - *Constrained Generation:* Concept coverage jumped from 44.0% to **67.0%** on ChatGPT and 15.0% to **45.0%** on GPT-4.
    - *Math Reasoning:* Modest gains on GSM-8k with internal feedback alone (+0.2% on GPT-4 to 93.1%) because models struggle to spot subtle math errors in their own code, but when paired with an external correctness oracle, accuracy improved by **+4.8% on GPT-3.5** and **+0.7% on GPT-4**.
  - **Analysis:** Structured multi-aspect feedback significantly outperformed generic feedback or no feedback, with the largest quality improvements occurring in the first 1 to 2 iterations.

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Would you like a comparative analysis focusing on how these architectures and reasoning frameworks build upon one another, such as how retrieval and iterative feedback address the factual limitations of large autoregressive models?